POLICY GRADIENT · SOCCER GAME

Policy gradient on the soccer game

REINFORCE, A2C and PPO, trained the standard way and by fictitious play, scored by how often the trained player wins when the opponent is the best response to it.

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1. Win rates against the best response

learner	Deterministic board (A10)			Random move-order board			Continuing game		
	ball seat	other seat	balanced	ball seat	other seat	balanced	ball seat	other seat	balanced
Exact solver	**0%**	**0%**	**0%**	**67%**	**39%**	**53%**	**0%**	**0%**	**0%**
REINFORCE, standard	0%	1%	0%	13%	0%	7%	0%	0%	0%
REINFORCE, fictitious play	2%	1%	2%	8%	0%	4%	0%	0%	0%
A2C, standard	2%	1%	2%	14%	1%	8%	0%	0%	0%
A2C, fictitious play	1%	2%	1%	12%	1%	7%	0%	0%	0%
PPO, standard	0%	0%	0%	9%	0%	5%	0%	0%	0%
PPO, fictitious play	0%	0%	0%	5%	0%	2%	0%	0%	0%
A2C, exact critic, standard	–	–	–	–	–	–	0%	0%	0%
A2C, exact critic, fictitious play	–	–	–	–	–	–	0%	0%	0%

- **Win rate**: the share of 1,000 games the trained player wins (it scores while the opponent does not; games with no goal or equal goals are ties), when the opponent plays the exact best response to it, the move that minimizes its expected discounted score difference. Mean of 3 seeds.
- **Ball seat**: the player that holds the ball at kickoff. **Other seat**: the same learner as the other player. **Balanced**: their 50/50 average, one number that does not depend on who starts with the ball. The exact solver is the top row, as context rather than a target.
- The random board's kickoff is worth +0.15 to the ball holder even against a perfect best response, so the exact solver wins more than half its games from that seat. On the deterministic board the exact solver ties itself; in the continuing game a win is more goals than the opponent in 100 steps. A dash means the learner was not run on that board.

2. Setup

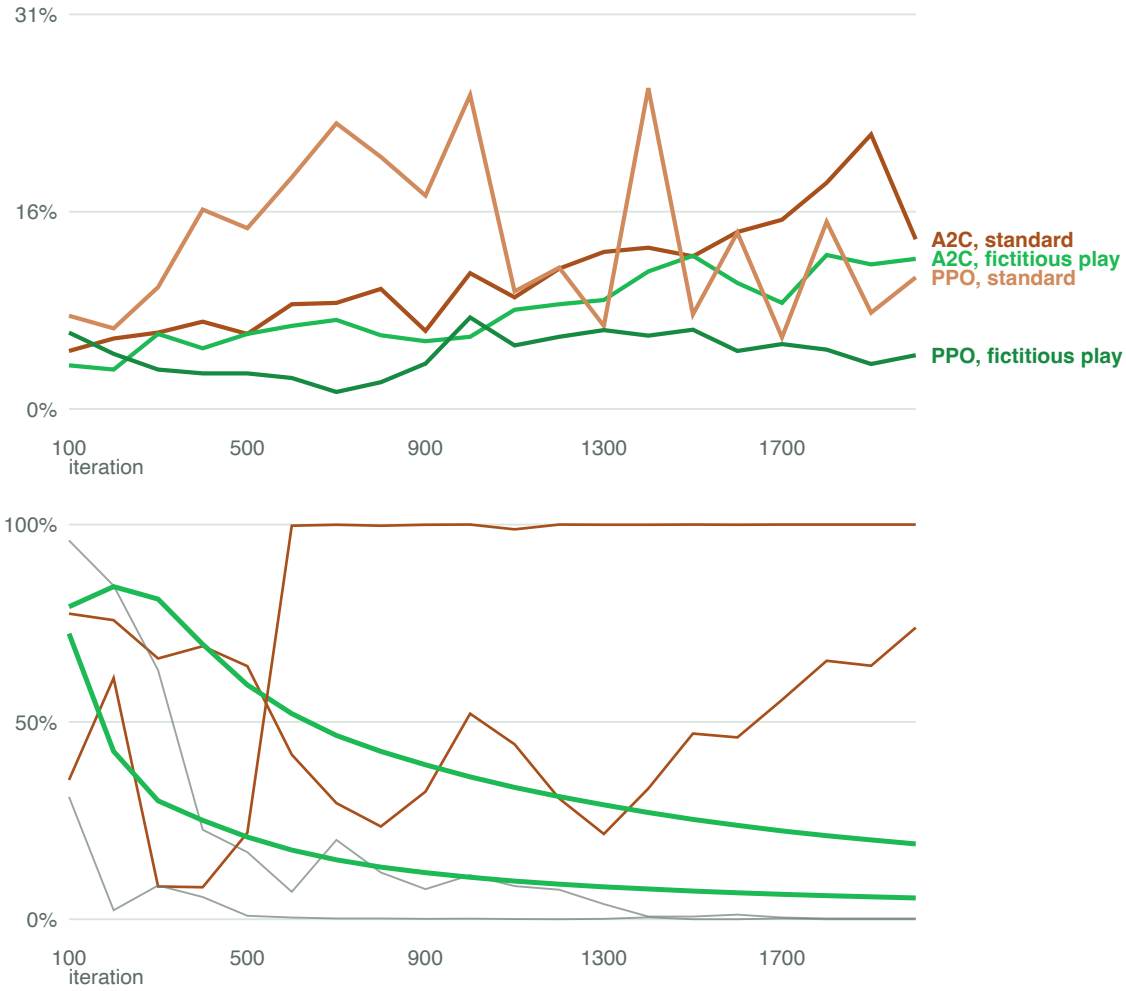
setting	value
Game	7×5 board, goal rows 1–3, fixed 100 steps, discount 0.9; the number of steps left is a network input. Three boards: deterministic (A10), random move order, and the continuing game (play restarts after a goal).
Algorithms	REINFORCE (discounted returns), A2C (10-step bootstrapped advantage; also with the exact solver's values as a frozen critic), PPO (GAE, clip 0.2, 4 epochs).
Updates	Each iteration: 64 episodes of 100 steps, one gradient step per player on all the data. 2,000 iterations, learning rate 10^{-3}, entropy bonus 0.01, 64×64 networks, a separate network per player.
Standard	Both current networks play each other and keep updating.
Fictitious play	Each player trains against a frozen snapshot of the other drawn from all its past snapshots (one every 5 iterations); it reports the average of its own snapshots. Policies, never weights, are averaged.

3. Findings

- **No learner replicates the exact solver.** On the random board the exact solver wins 67% from the ball seat, 39% from the other seat and 53% balanced; the six learners win 5% to 14%, 0% to 1% and 2% to 8%. On the deterministic board the exact solver ties itself and the learners win 2% or less; in the continuing game every learner wins 0%.
- Against a random player the learners win 61% to 96% of games (deterministic board); fictitious play does better than standard training.
- Starting from the 94 states where the exact answer has to mix, the learners win 11% to 15% against the exact Nash policy; the exact solver wins 50%.
- **Best settings, on 10 seeds:** A2C fictitious play at 10,000 iterations wins 30% from the ball seat (95% interval 24% to 36%) against 9% at 2,000; PPO with replayed data wins 28% (13% to 44%) against 10%. Both gains hold up and stay well below the exact solver.

4. Does training converge?

Random board. Every 100 iterations the policy each learner would report is scored against the exact best response (mean of 3 seeds), and the probability it gives to the exact solver's kickoff move (R; the exact solver plays it 100%) is recorded for seed 0. Orange: standard training's current network. Grey: the current network under fictitious play. Green: fictitious play's average.

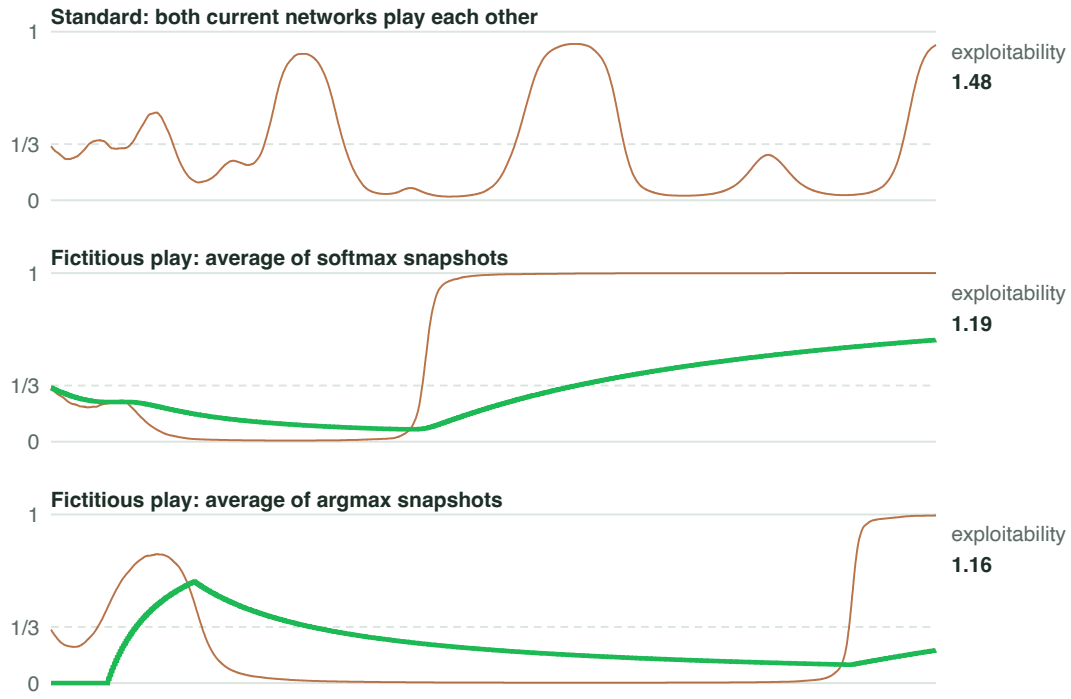


In the second half of training the win rate against the best response stays within A2C standard 9% to 22%; A2C fictitious play 8% to 12%; PPO standard 6% to 25%; PPO fictitious play 4% to 6%; the exact solver wins 67% from the ball seat. Standard training keeps swinging; the fictitious-play average is steadier but flat and low.

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5. Rock-paper-scissors with neural networks

Each player is a small softmax network trained by REINFORCE (3,000 iterations, 64 games each, 3 seeds; seed 0 shown). Orange: the current network's share of rock. Green: the average over past snapshots (fictitious play), of softmax policies or of argmax (pure) policies. Exploitability (right) is 0 at one third each. With the soccer settings the standard networks cycle (1.48) and neither average reaches one third (softmax 1.19, argmax 1.16). With a much larger entropy bonus (0.2) and plain gradient steps all three settle: exploitability 0.10 standard, 0.07 softmax average, 0.28 argmax average. That is a stabilization effect of the changed objective, not evidence of convergence to the unregularized equilibrium.



6. Training choices

Each comparison changes one thing from the usual setting (2,000 iterations of 64 episodes, a snapshot every 5 iterations, all 100 steps in the loss) on 3 seeds. Changes are in percentage points; single seeds of one setting differ by 10 to 30 points, so differences of that size are noise.

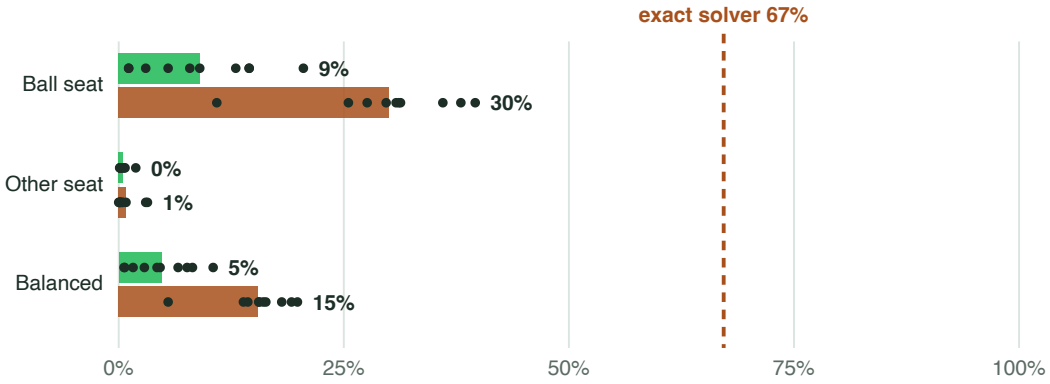
- **One shared network** (two output heads), random board: –3 to +11 points.
- **Last 10 of 100 steps left out**, random board: –8 to +4 points; PPO with standard training stops scoring.
- **Averaging argmax (pure) snapshots** instead of softmax ones: –6 to –4 points on the random board; no gain on the other boards.
- **Entropy bonus 0.2 instead of 0.01** (an ablation; it changes the optimization problem, so it does not show convergence to the unregularized equilibrium), random board: REINFORCE –4, A2C –6, PPO +24 points.
- **Leaving out the last steps: how many?**: win rate, change in points from "all steps": last 5 left out –10 to +11; last 10 left out –8 to +4; last 20 left out –9 to +11; last 40 left out –10 to +13.
- **Leaving out the last steps in the continuing game**: share tied with the exact Nash policy, change in points from "all steps": last 5 left out –59 to –43; last 10 left out –67 to –15; last 20 left out –66 to –44; last 40 left out –40 to –33.
- **How long to train between snapshots**: win rate, change in points from "snapshot every 5 iterations (used elsewhere)": every iteration –6 to –2; every 25 –11 to –3; every 100 –10 to +4.
- **Five times as many iterations**: win rate, change in points from "2,000 iterations": 10,000 iterations –9 to +23.
- **Mini-batches and replayed data**: win rate, change in points from "all data, one step": mini-batches of 256 –11 to +6; replay of 4 iterations –2 to +37.

- **The exact solver's values as the critic: random board**: win rate, change in points from "learned critic": exact critic –8 to –5.
- **The exact solver's values as the critic: deterministic board**: share tied with the exact Nash policy, change in points from "learned critic": exact critic –43 to +0.
- **Averaging weights instead of policies**: win rate, change in points from "average of policies": average of weights –7 to –3; average of the output layer's weights –4 to +1.

Ten seeds for the two best-looking settings

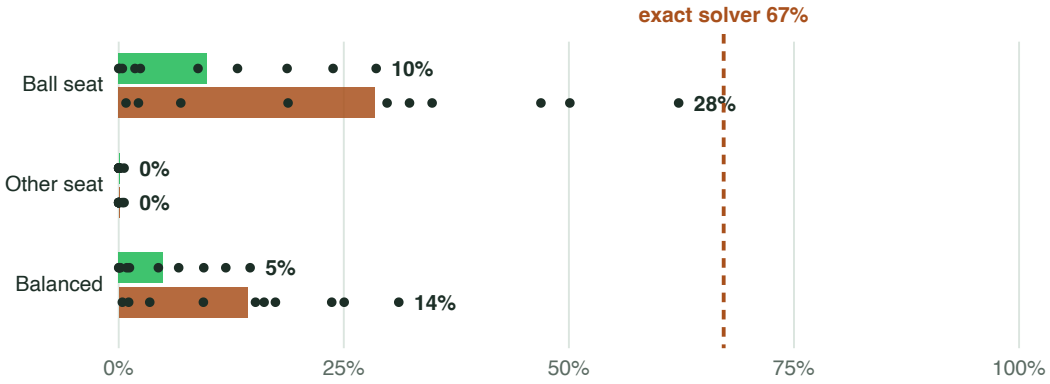
Seeds 0 to 9, random board; 95% intervals (Student's t) in brackets; green: usual setting, orange: changed setting, dots: single seeds. The exact solver wins 67% from the ball seat, 39% from the other seat and 53% balanced.

A2C, fictitious play: 10,000 iterations



Ball seat 9% (4% to 14%) → 30% (24% to 36%), a change of +21 points (+14 to +28); balanced 5% → 15%, +11 points (+7 to +14). The interval for the change excludes zero.

PPO, standard: replay of 4 iterations



Ball seat 10% (2% to 17%) → 28% (13% to 44%), a change of +19 points (+3 to +35); balanced 5% → 14%, +9 points (+1 to +17). The interval for the change excludes zero.

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7. Related work

Has fictitious play been combined with policy gradient before? Yes, in several forms, and the convergence theory is much older than the neural networks. I checked the titles, authors and venues against Google Scholar (its top result for fictitious play, policy gradient and zero-sum Markov games

was Liu, Yang and Wang), the publishers' and proceedings pages, and arXiv. JSTOR's search page would not load, so the 1951 papers are cited from the linked secondary source. The one-line summaries follow each paper's abstract or listing and I have not checked the proofs. I did not find a result that proves convergence for deep policy gradient with fictitious play in zero-sum Markov games; the closest results are tabular, regularized or for special game classes.

Older work

- Iterative solution of games by fictitious play (Brown, 1951; Activity Analysis of Production and Allocation). Introduced fictitious play as a computer method for finding equilibria of two-player zero-sum games (first proposed in 1949). Not read directly; described in the linked paper.
- An iterative method of solving a game (Robinson, 1951; Annals of Mathematics 54(2)). Proved Brown's conjecture that fictitious play converges in two-player zero-sum games; Shapley (1964) later showed that convergence fails in general games. Not read directly; described in the linked paper.
- Markov games as a framework for multi-agent reinforcement learning (Littman, 1994; ICML, pp. 157-163). Two-player zero-sum Markov games and minimax-Q, a Q-learning variant that replaces the max with a minimax over the joint action (each update solves a linear program).
- Fictitious play property for games with identical interests (Monderer and Shapley, 1996; Journal of Economic Theory 68(1), pp. 258-265). Defines n-person games with identical interests (best-response equivalent in mixed strategies to a game with identical payoff functions) and proves that every such game has the fictitious-play property. Abstract read on a bibliographic mirror; the publisher's page was blocked.
- Potential games (Monderer and Shapley, 1996; Games and Economic Behavior 14(1)). Defines several notions of potential function for games in strategic form and characterizes the games that have one: all players' incentives to change strategy are captured by a single function.
- Multiagent learning using a variable learning rate (Bowling and Veloso, 2002; Artificial Intelligence 136(2)). Gradient-style learners in stochastic games: the WoLF (win or learn fast) learning-rate rule, with a convergence proof in self-play for a restricted class of matrix games.
- On the global convergence of stochastic fictitious play (Hofbauer and Sandholm, 2002; Econometrica 70(6)). Global convergence of stochastic (perturbed) fictitious play for several classes of games, including zero-sum and potential games.
- Nash Q-learning for general-sum stochastic games (Hu and Wellman, 2003; Journal of Machine Learning Research 4). Q-learning toward Nash equilibria of general-sum Markov games; convergence is proved only under restrictions on the stage games that arise during learning.
- Generalised weakened fictitious play (Leslie and Collins, 2006; Games and Economic Behavior 56(2)). A broad class of adaptive learning processes, including an actor-critic process that learns only from rewards, shown to converge in games with the fictitious-play property.
- Brown's original fictitious play (Berger, 2007; Journal of Economic Theory 135(1)). Brown's 1951 process updates beliefs in a different order from the version used today; with that order, a short convergence proof for nondegenerate ordinal potential games.

Recent work

- Fictitious Self-Play in Extensive-Form Games (Heinrich, Lanctot and Silver, 2015; ICML, PMLR 37). Fictitious play in behavioural strategies of extensive-form games, and a sample-based version

(fictitious self-play); approximate Nash equilibria in poker games.

- Deep Reinforcement Learning from Self-Play in Imperfect-Information Games (Heinrich and Silver, 2016; arXiv). Neural fictitious self-play (NFSP) learns approximate Nash equilibria of poker games end to end; ordinary reinforcement learning diverged on the small game (Leduc poker) where NFSP moved toward equilibrium.
- Independent Policy Gradient Methods for Competitive Reinforcement Learning (Daskalakis, Foster and Golowich, 2020; NeurIPS 33). Both players running policy gradient simultaneously reach a min-max equilibrium of a two-player zero-sum stochastic game, provided the learning rates follow a two-timescale rule (which the authors show is necessary).
- Policy Optimization in Zero-Sum Markov Games: Fictitious Self-Play Provably Attains Nash Equilibria (Liu, Yang and Wang, 2020; OpenReview). A smooth fictitious self-play whose best-response step is a PPO step with entropy regularization reaches a neighbourhood of a Nash equilibrium at a sublinear rate in zero-sum Markov games (a guarantee for the regularized method, not the exact equilibrium).
- Provably Efficient Fictitious Play Policy Optimization for Zero-Sum Markov Games with Structured Transitions (Qiu, Wei, Ye, Wang and Yang, 2021; ICML, PMLR 139). Fictitious-play policy optimization with optimism for zero-sum Markov games with structured but unknown transitions, with square-root-of-episodes regret bounds.
- Policy Optimization for Markov Games: Unified Framework and Faster Convergence (Zhang, Liu, Wang, Xiong, Li and Bai, 2022; NeurIPS 35). Each iteration updates the policy at every state with a matrix-game algorithm and then the values; the state-wise average policy converges to an approximate Nash equilibrium of two-player zero-sum Markov games.
- Fictitious play in zero-sum stochastic games (Sayin, Parise and Ozdaglar, 2022; SIAM Journal on Control and Optimization). A fictitious-play variant that combines classical fictitious play with Q-learning for two-player zero-sum stochastic games.
- Fictitious Play in Markov Games with Single Controller (Sayin, Zhang and Ozdaglar, 2022; ACM EC). A variant of fictitious play converges in identical-interest Markov games where one player controls the transitions; together with earlier results it covers both zero-sum and identical-interest Markov games.
- Fast Policy Extragradient Methods for Competitive Games with Entropy Regularization (Cen, Wei and Chi, 2024; Journal of Machine Learning Research 25(4)). Entropy regularization gives linear-rate convergence to the regularized equilibrium (including in zero-sum Markov games), and tuning the weight approaches an approximate Nash equilibrium of the unregularized game.